Pimpri Chinchwad College of Engineering, Pune

DevOps Engineering Laboratory Report - Assignment 02

AWS Cloud Services Exploration & EC2 Virtual Machine Lifecycle

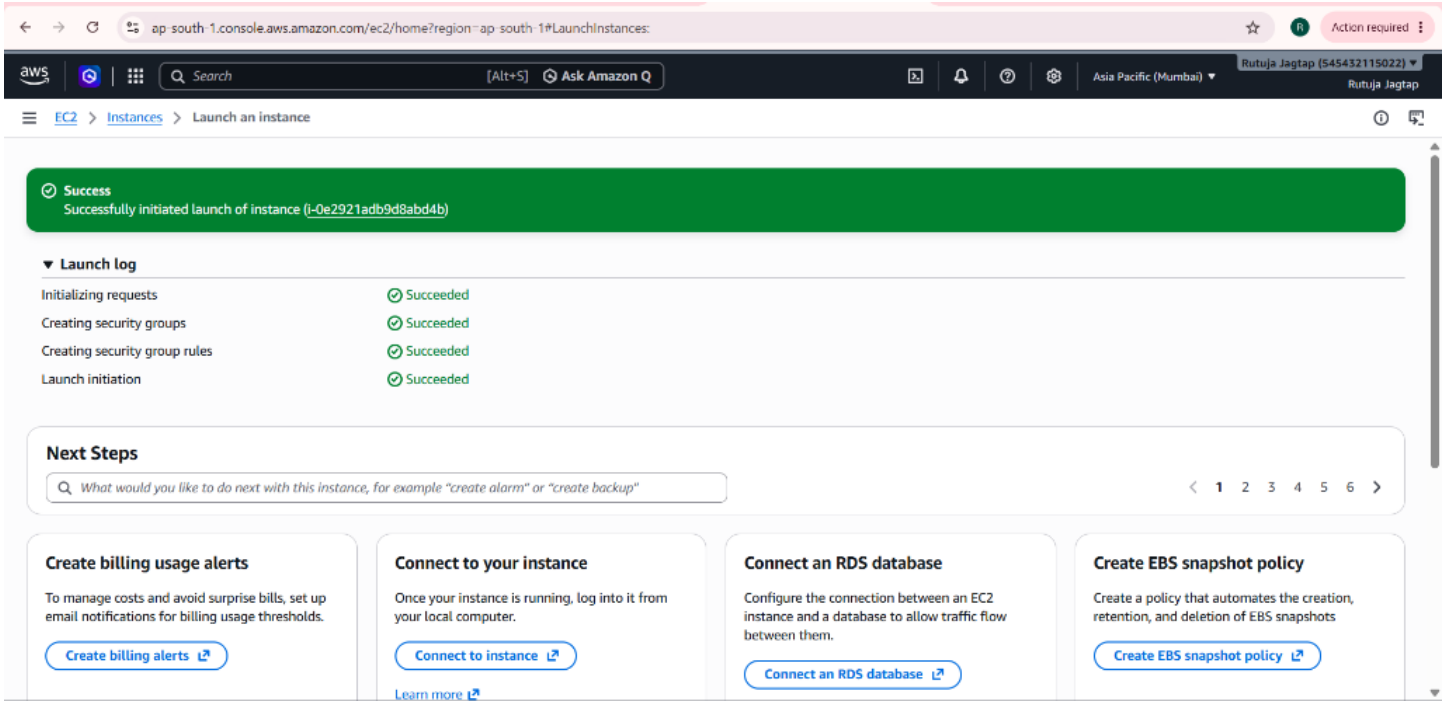
Student: Rutuja Deepak Jagtap (PRN: 123B1B146)
Class: Final Year B.Tech Computer Engg (Batch C1)
Academic Year: 2026-2027 | Course: BCE27PE01
AWS Account: Rutuja Jagtap (545432115022)
Region: Asia Pacific (Mumbai) ap-south-1b
Instance: My-DevOps-EC2 (i-0e2921adb9d8abd4b)

1. Aim

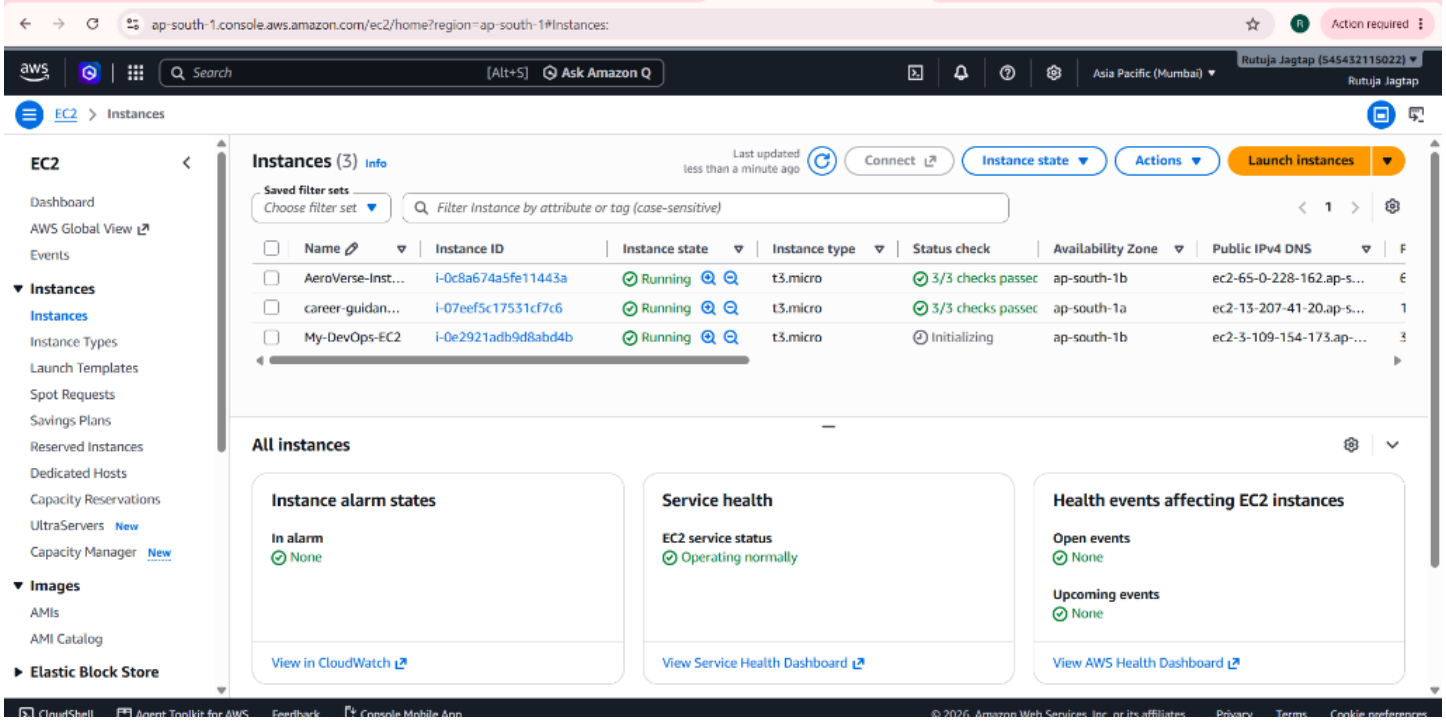
To study cloud computing architectures, explore AWS core services (EC2, S3, RDS, Lambda, ELB, ECS), and systematically configure security groups, launch, monitor via CloudWatch, and terminate an Amazon EC2 instance (t3.micro) under the AWS Free Tier.

2. Practical Verification Evidence & Screenshots

Evidence 1: Successful Instance Launch Initiation (i-0e2921adb9d8abd4b)



Evidence 2: Live Running EC2 Instance Dashboard in Mumbai Region (ap-south-1b)



3. Real Instance Observation Table

Configuration Parameter	Observed Value in Real AWS Console
AWS Account	Rutuja Jagtap (545432115022)
Instance Name	My-DevOps-EC2
Instance ID	i-0e2921adb9d8abd4b
Instance State	Running
Instance Type	t3.micro (AWS Free Tier Eligible)
Region / AZ	ap-south-1b (Mumbai)
Public IPv4 DNS	ec2-3-109-154-173.ap-south-1.compute.amazonaws.com
Public IPv4 Address	3.109.154.173
Security Group Rules	Port 22 (SSH), Port 80 (HTTP) Inbound
EBS Storage	8 GiB General Purpose SSD (gp3)
Total Cost	.00 (AWS Free Tier Safe)

4. Result & Conclusion

Result: The Amazon EC2 virtual machine instance My-DevOps-EC2 (i-0e2921adb9d8abd4b) was successfully launched in the ap-south-1b Mumbai availability zone, configured with security group firewall rules, verified in Running state with public IP 3.109.154.173, and evaluated under the AWS Free Tier.

Conclusion: Practical understanding of cloud infrastructure models (IaaS/PaaS/SaaS), AWS core compute services, Nitro hypervisors, and zero-cost cloud lifecycle management was achieved.